

Taehyung Noh

yestaehyung@hanyang.ac.kr |

 taehyung-noh |  yestaehyung |

521, Fusion Technology Center, Hanyang University, Seoul, South Korea 04763

RESEARCH INTERESTS

My research interest lies in Human-AI Interaction, with a focus on developing trustworthy and personalized AI systems.

EXPERIENCE

- **Hanyang Human-Centered Computing Laboratory**  Jan. 2022 - Present
Research Assistant Seoul, Korea
- **Ajou Human-Computer-Interaction Lab at Dept. of Software** Aug. 2020 - Dec. 2021
Undergraduate Research Assistant Suwon, Korea

EDUCATION

- **Hanyang University** Mar. 2022 - Present
M.S. & Ph.D. Integrated Student, Department of Artificial Intelligence (Advisor: Kyungsik Han) Seoul, Korea
- **Ajou University** Mar. 2018 - Feb. 2022
Bachelor of Engineering, Department of Software Suwon, Korea

PROJECTS

- **KETI Industrial AI Data Preprocessing Platform** Present
 - Developing an AI data preprocessing platform for industrial applications.
 - Providing an integrated solution for efficiently processing and analyzing data across various industrial domains.
- **TRIPLE (AAAI)** 2025
First Author
 - TRIPLE (Theory-guided Reasoning for Intent and habIt Profiling with LLMs for pErsonalization) is a novel framework that integrates dual-process theory and the Theory of Planned Behavior (TPB) into LLM-based user modeling.
 - Constructs both habitual and intentional behavior profiles, then generates behavioral rationale that explains the interaction between these processes to predict user behavior.
- **TRIPLE (CIKM)** 2025
First Author
 - TRIPLE is a profiling technology that combines the Theory of Planned Behavior (TPB) with Large Language Models (LLMs).
 - Uses LLMs to understand a user's psychological motivations and refines their profile by comparing predictions with actual behavior, dramatically improving personalization services.
- **Red Teaming Attack Dataset Analysis** 2024
Project Leader
 - Analyzed real-world LLM vulnerabilities using attack data from the 2023 DEF CON 31 Generative AI Red Teaming Challenge.
 - Preprocessed and relabeled the dataset to identify which target categories are most susceptible to harmful-content attacks.
 - Classified victim groups and categorized prompt types to reveal systematic patterns in successful jailbreaks.
- **Multi-Agent Personality Detection System (PADO)** 2024
Second Author
 - PADO is a multi-agent system that detects personality traits (OCEAN) from user-generated text.
 - Multiple specialized agents collaborate to perform more accurate personality analysis.
- **Fashion-FINE** 2024
Third Author
 - Fashion-FINE is a Vision-Language Pre-training model for fine-grained fashion retrieval.

- Introduces modality-agnostic adapter, hard negative mining with focal loss, and comprehensive cross-modal alignment.

• My Own Style (MOS)

First Author

May. 2022 - Present

- Investigated how domain expertise shapes user perception and satisfaction with fashion-recommendation outcomes.
- Built MOS dashboard that analyzes outfit images and recommends similar/diverse fashion items.
- Large-scale study with 166 participants showing fashion experts prefer highly similar recommendations while non-experts prefer diverse results.

PUBLICATIONS

Conference Papers

- [C.1] **Taehyung Noh**, Seungwan Jin, Haein Yeo, Kyungsik Han (2026). TRIPLE: Theory-Driven Integration of Planned and Habitual Behaviors for LLM-based Personalization. The AAAI Conference on Artificial Intelligence (**AAAI 2026**)
- [C.2] **Taehyung Noh**, Seungwan Jin, Haein Yeo, Kyungsik Han (2025). [Externalizing Social-Cognitive Structures for User Modeling: Toward Theory-Driven Profiling with LLMs](#). The ACM International Conference on Information and Knowledge Management (**CIKM 2025 Short**)
- [C.3] Haein Yeo, **Taehyung Noh**, Seungwan Jin, Kyungsik Han (2025). [PADO: Personality-induced multi-Agents for Detecting OCEAN in human-generated texts](#). The International Conference on Computational Linguistics (**COLING 2025**)
- [C.4] Seungwan Jin, Hoyoung Choi, **Taehyung Noh**, Kyungsik Han (2024). [Integration of global and local representations for fine-grained cross-modal alignment](#). The European Conference on Computer Vision (**ECCV 2024**)
- [C.5] **Taehyung Noh**, Haein Yeo, Myungjin Kim, Kyungsik Han (2023). [A study on user perception and experience differences in recommendation results by domain expertise: the case of fashion domains](#). The ACM Conference on Human Factors in Computing Systems Late-Breaking Work (**CHI LBW 2023**)

Journal Papers

- [J.1] Haein Yeo, **Taehyung Noh**, Kyungsik Han (2025). LLM-Generated Content-Based Explanations for User Experience in Fashion Recommender Systems. **Fashion and Textiles**

Technical Reports

- [R.1] Yeajin Shin, Kyungsik Han, **Taehyung Noh**, Mingon Jeong (2025). [LLM Attack Strategies](#). **TTA Report**

PATENTS

- [P.1] Seungwan Jin, Kyungsik Han, **Taehyung Noh**, Junghyun Kim (2025). Dynamic Persona-based Prediction System and Method for Personalized Long-term Mental Health Monitoring.
- [P.2] Kyungsik Han, **Taehyung Noh** (2025). Method and Device for Generating User Profiles Using Artificial Intelligence.
- [P.3] Bogoan Kim, Dayoung Jung, **Taehyung Noh**, Kyungsik Han (2023). Visualization Analysis Tool Equipped with VR Multimodal Sensor Data Processing Pipeline Collected from People with Autism Spectrum Disorder.
- [P.4] Kyungsik Han, **Taehyung Noh**, Haein Yeo, Myungjin Kim (2023). Device and Method for Providing Dashboard Services on Fashion Image Analysis.
- [P.5] Kyungsik Han, **Taehyung Noh**, Haein Yeo, Myungjin Kim (2023). Device and Method for Similar Image Recommendation Using Fashion Attributes and Image.
- [P.6] Kanghwi Ahn, Junseok Ryu, Kyungsik Han, **Taehyung Noh** (2023). Method for Analyzing Image and Text-based Characteristics of Objects.
- [P.7] Kyungsik Han, Seungwan Jin, Hoyoung Choi, **Taehyung Noh** (2022). Image-Text Retrieval Model Construction Method and Service Device.

HONORS AND AWARDS

Best Paper Award , Korea Data Mining Society	Nov. 2024
Best Presentation Award , Korea Software Congress (KSC 2023)	Dec. 2023
Best Inventor Award , Seoul International Invention Fair	Dec. 2022
Participation Award , mySUNI Creative Challenge	Dec. 2022

TEACHING EXPERIENCE

Co-lecturer

- ITE3062: Human-Computer Interaction (In English) Fall 2024

Teaching Assistant (TA)

- SOI3010: Laboratory practice of Intelligence Computing 2 Fall 2024

ACADEMIC SERVICES

Conference Reviewer

- Conference on Empirical Methods in Natural Language Processing (EMNLP) 2024
- ACM Conference on Human Factors in Computing Systems (CHI) 2024
- ACM International Conference on Information and Knowledge Management (CIKM) 2023

REFERENCES

1. **Kyungsik Han**
Associate Professor, Hanyang University
Department of Artificial Intelligence
kyungsikhan@hanyang.ac.kr